Stormscope – General data

1. General

The Stormscope (WX–500) is a passive sensor that uses a receive antenna to detect the bearing and range of electromagnetic signals (lightning).

2. Description

A. Stormscope (WX–500)

The Stormscope (WX–500) System maps electromagnetic signals (lightning) 360 degrees around the aircraft and to a maximum radius of 200 nautical miles.

When selected in the horizontal situation indicator (HSI) overlay, the partial compass HSI format displays information from the Lightning Sensor System. There are two components of the lightning display on the HSI, mode/fault annunciations/Strike Rate and lightning cell strike data. Mode/fault and Strike Rate annunciations are placed outside of the HSI Display area while lightning strike signals are placed inside the HSI. LX CLR in the HSI overlay clears all strike signals that have been received.

The Stormscope system consists of a receiver installed in the rear avionics rack and a top mounted antenna installed at FS400. A receiver inhibit line is interfaced to the aircraft audio control panel ensuring that erroneous strike signals are not displayed when a radio transmitter is keyed.

The WX–500 contains three levels of self test. Power-on self test, continuous self test and pilot initiated self test. The results are communicated to the host through System Status messages and Faults Code messages. Pilot initiated self test takes approximately 30 seconds during which time data is not updated.

28 Volt DC power is protected by a 3 Amp STORMSCOPE (M4) (Right Bus) circuit breaker located on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

The Stormscope (WX–500) interfaces to the following aircraft systems:

1) Modular Avionics Unit (MAU) (refer to 31-43-00, Modular Avionics Unit (MAU) (Mod 6/2049) – General Data).

2) Audio Control Panel System (refer to 23-53-00, Audio Control Panel – General Data).

3. Adjustment/Test

A. Function Test Stormscope (WX–500)

1) General

These procedures are for the function test of the Stormscope (WX–500).

2) Function Test

a) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

b) Wait several minutes for the aircraft electrical and avionics systems to initialize.

c) Ensure that STORMSCOPE (M4) (Right Bus) circuit breaker on the avionics circuit breaker panel is closed.

d) On the lower MFD, select the WX/TAWS/LX page, then select LX.

e) Press the bezel button next to LX SELF TEST and make sure that the HSI LX MODE shows a white TEST message.

f) Ensure that the HSI LX MODE shows a white STRIKE message after the self test is complete.

g) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

4. Removal/Installation

WARNING **INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND INDIRECT EFFECTS OF LIGHTNING (IEL). ENSURE INTEGRITY OF THE HIRF/IEL PROTECTION FEATURES ARE MAINTAINED IN ACCORDANCE WITH THE FOLLOWING:**

CHASSIS GROUNDING – ALL RACK MOUNT / REMOTE MOUNT UNITS SHALL BE ELECTRICALLY BONDED TO THE AIRFRAME. ENSURE THAT THE MATING SURFACES BETWEEN THE LRU MOUNTING TRAY (OR LRU MOUNTING FEET IF A TRAY IS NOT USED) PROVIDE A LOW IMPEDANCE ELECTRICAL PATH. ALL MATING SURFACES MUST BE FREE OF ALL PAINT AND OTHER NONCONDUCTIVE ELEMENTS. REFER TO BONDING REQUIREMENTS IN 20-75-00, ELECTRICAL BONDING AND GROUNDING OF AIRCRAFT STRUCTURES – GENERAL DATA.

SHIELD GROUNDS – VERIFY SHIELD GROUNDING AT CONNECTOR BACKSHELLS AS PER REQUIREMENTS IN PSM 1-64-2W.

BACKSHELLS OR STRAIN RELIEFS – VERIFY PROPER ELECTRICAL BONDING BETWEEN THE BACKSHELL/STRAIN RELIEF AND THE MOUNTING TRAY OR LRU CASE AS PER REQUIREMENTS IN PSM 1-64-2W.

A. Removal and Installation of Stormscope (WX–500)

1) General

These procedures are for the removal and installation of the Stormscope (WX–500).

2) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Pull the STORMSCOPE (M4) circuit breaker on the avionics circuit breaker panel and fit safety clip.

c) Gain access to the rear avionics shelves through the rear baggage compartment.

d) Disconnect the electrical connector on the Stormscope unit.

e) Loosen retaining screw clamp from the front and slide the unit out of the mounting tray.

f) Remove unit from aircraft.

g) Visually inspect mounting tray. If damaged replace as per Para 4.C.

Note Stow connectors of the removed components.

3) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the STORMSCOPE (M4) circuit breaker is pulled and fitted with safety clip.

c) Slide the unit onto the mounting tray. Tighten the retaining screw clamp at the front lip of the STORMSCOPE unit.

d) Reconnect the electrical connector to the Stormscope unit. Ensure connections are secure.

e) Remove safety clip and close STORMSCOPE (M4) circuit breaker on the avionics circuit breaker panel.

f) Test the Stormscope as per Para 3.A.2).

B. Stormscope Antenna (NY–163)

1) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Pull the STORMSCOPE (M4) circuit breaker on the avionics circuit breaker panel and fit safety clip.

c) Carefully cut the sealant from around the base of the antenna.

d) Remove the two screws from the base of the antenna.

e) Carefully lift the antenna clear of the fuselage until there is access to the coaxial connection.

f) Disconnect the coaxial connector.

Note Stow connectors of the removed components.

2) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the STORMSCOPE (M4) circuit breaker is pulled and fitted with safety clip.

c) Solvent clean the electrical contact bond areas immediately before the installation (refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

d) Bring the antenna assembly close to the fuselage, connect the coaxial connector.

e) Secure the antenna assembly on fuselage with the two mounting screws.

f) Solvent clean the exposed area of the airframe around the antenna (refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

g) Touch up the exposed area of airframe around the antenna if required with F19 primer by brush within 24 hours of the antenna installation. Ensure that the primer re-finishing provides complete coverage of the affected area.

h) If the affected area requires paint touch up, brush coat the area with the top-coat paint system matching the existing.

i) Use MIL–PRF–81733 sealant to both fillet seal around the antenna base and seal the securing screws. The aircraft will be ready to fly approximately 24 hours after sealing.

j) Remove safety clip and close STORMSCOPE (M4) circuit breaker on the avionics circuit breaker panel.

k) Check the antenna electrical bonding according to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data.

l) Test the Stormscope system as per Para 3.A.2).

C. Removal and Installation of Stormscope (WX–500) Mounting Tray

1) General

These procedures are for the removal and installation of the Stormscope mounting tray.

2) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Gain access to the rear avionics shelves through the rear baggage compartment.

c) Loosen retaining screw clamp from the front lip of the transceiver and slide the unit out of the tray mount.

d) Remove the four screws that secure the mounting tray to the bottom of the avionics rack.

e) Remove mounting tray from aircraft.

3) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Align the tray to the mounting holes on the rear avionics rack.

c) Install the four screws that secure the mounting tray to the avionics rack.

5. Troubleshooting

A. Stormscope (WX–500)

1) If the Stormscope (WX–500) is deemed inoperative check the following:

a) Ensure STORMSCOPE (M4) circuit breaker is closed.

b) Check that the unit is securely installed and the retaining mechanism is securely tightened.

c) Visually inspect Stormscope connector for damaged wires or contacts.

d) If troubleshooting at the wire level is required refer to the applicable wiring diagrams in Wiring Manual PSM1-64-2W.

e) If repair or service is required, send to appropriate rated avionics repair facility.